

Lab 9 Solutions(Grp 9-16)

Sam got a new Challenge

```
def isInteger(x):
    try :
        int(x);
        return True
    except ValueError:
        return False

n = int(input())
ans = ""
for i in range(n):
    x = input()
    if isInteger(x):
        ans += "Integer"
    else :
        ans += "Not an Integer"
    ans += "\n"
print(ans)
```

Reduce fractions :

```
inp = input().split(" ")
a,b = int(inp[0]),int(inp[1])

def isPrime(n):
    # function to check if a number is prime or not.
    for i in range(2,n):
        if n % i ==0 :
            return False;
    return True;

def getPrimeDivisors(n):
    #function to get prime divisors of a number along with their count
    prime_divisors = {}
    upto = n+1;
    for i in range(2,upto):
```

```

    if isPrime(i):
        while n % i == 0 :
            if i not in prime_divisors.keys():
                prime_divisors[i]=0
            prime_divisors[i]+=1
            n/=i

    return prime_divisors

a_divisors = getPrimeDivisors(a)
b_divisors = getPrimeDivisors(b)
# now find all common divisors of a and b.
common_divisor = {}
for key in a_divisors.keys():
    if key in b_divisors.keys():
        common_divisor[key] = min(a_divisors[key],b_divisors[key])
# now remove all common divisors from a and b.

for key in common_divisor.keys():
    while common_divisor[key] > 0:
        a//=key
        b//=key
        common_divisor[key]-=1
print(a,b)

```

Reciprocity ;

```
raf, age = [int(x) for x in input().split()]
```

```

try:
    ras = 1/(raf-age)
    print(f'{ras:.2f}')

```

```

except:
    print("Ambivert")

```

Age assertion :

```
n = int(input())
```

```
try:
    assert n>=0
    print(n)

except:
    print("Age can't be negative")
```

Tribonacci sequence :

```
n = int(input())

try:
    a1 = 0
    a2 = 1
    a3 = 1

    if (n==1):
        print(a1)
    elif(n==2):
        print(a2)
    elif (n==3):
        print(a3)
    else:
        ## none of the base cases, then proceeding further
        for i in range(4, n+1):
            tribo_sum = a1+a2+a3
            a1=a2
            a2=a3
            a3=tribo_sum

        print(tribo_sum)

except:
    print("Not allowed")
```

Target word ;

```
a = input()
l = [int(r) for r in input().split()]
```

```
res = ""
for i in l:
    try:
        res += a[i]
    except Exception:
        res += "_"
print(res)
```

Merge lectures :

```
n=int(input())
fl=[]
fl.append([int(i) for i in input().split()])
for i in range(n-1):
    l1 = [int(i) for i in input().split()]
    if(l1[0]==fl[len(fl)-1][0]):
        fl[len(fl)-1].append(l1[1])
    else:
        fl.append(l1)
print([tuple(i) for i in fl])
```